



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Yellow Belt (CLSSYB)

Course Code: C524

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release — CLSSYB 2-day lesson plan aligned to the 10 hands-on labs.	Dr. Alfred Ang
2	15 June 2026	Refreshed labs against the CSSC Yellow Belt reference.	Dr. Alfred Ang
3	1 July 2026	Aligned the schedule with the Contoso Service Desk case-study scenario.	Dr. Alfred Ang
4	23 September 2026	Major content expansion: course restructured to follow the DMAIC roadmap end to end; Lean Six Sigma concepts imported from the reference trainer deck and the CSSC body of knowledge and explained visually; 4 elective labs added (Affinity/Kano, VSM/Takt, Solution Matrix/Benchmarking/ FMEA, Descriptive Statistics/Implementation Plan) and Lab 4 (PDCA charter) reclassified as elective, giving 9 core and 5 elective labs; interactive problem-solving tools (5 Whys, Fishbone, collaborative Pareto, NovaSPC) integrated into the Analyze labs; slide references added to the schedule.	Dr. Alfred Ang
5	23 September 2026	Integrated the SIPOC & Process Map Builder (alfredang.github.io/sipoc/) into Lab 3 and elective Lab 12, with a tool walkthrough added to the Define phase and the online toolkit slide updated to list all five tools.	Dr. Alfred Ang
6	23 September 2026	Fixed clipped headings: slide titles now auto-fit to a single line so long lab titles	Dr. Alfred Ang

		can no longer overprint the LAB/ELECTIVE chips beneath them.	
7	20 July 2026	Table of contents restored to a live, updatable Word field (previously flattened to static text), so it refreshes on open or F9 while still rendering correctly in the distributed PDF.	Dr. Alfred Ang
9	23 September 2026	Labs reissued as ACTIVITY PACKS in the Tertiary Infotech house format: one folder per activity named "NN - Title", each holding a Facilitator Guide, a Learner Worksheet and a Checklist (DOCX + PDF) alongside its own mock data (data/*.csv) and blank worksheets (templates/*.csv). The previous labs/ Markdown layout is superseded and archived.	Dr. Alfred Ang

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Course Information

Course Title	Certified Lean Six Sigma Yellow Belt (CLSSYB)
Course Reference	C524
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch break)
Mode	Instructor-led, hands-on Lean Six Sigma activities using the Contoso Service Desk improvement scenario
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.
- LO2: Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.
- LO3: Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.
- LO4: Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.
- LO5: Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.
- LO6: Recommend improvement and control actions to sustain gains, and prepare for certification.

Training Resources and Activity Data

Each activity is a self-contained folder under activities/ — activities/NN - <Name>/ — holding the Facilitator Guide, the Learner Worksheet and the Checklist (each as DOCX and PDF), the mock data the learners analyse (data/*.csv) and the blank worksheets they complete (templates/*.csv).

- Shared data set: two weeks of Contoso Service Desk activity — 400 tickets, 96 defective, 120 defects, 6 opportunities per ticket.
- Baseline performance: yield 76%, DPU 0.30, DPMO 50,000, sigma level ~3.1; mean assignment time 55 min (median 46) against a 30-minute goal.
- Trainer note: the Activity 7 worked answers are in activities/07 - Pareto*/solution/ — do not release before learners attempt them.

- Learners need a spreadsheet application (Excel, Google Sheets or LibreOffice Calc) and a browser.

Course Schedule

Day 1 — Six Sigma foundations, Define phase and process mapping

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and course overview [slides 2–9]
10:00–11:15	75 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma; sigma levels and DPMO; $Y = f(X)$; belt roles; the DMAIC roadmap; DMAIC vs PDCA [slides 10–41]
11:15–12:15	60 min	Hands-on: Activity 1: Yellow Belt Role, Certification Paths, and Improvement Scenario
12:15–12:45	30 min	Lunch break
12:45–14:15	90 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; the project charter; problem and goal statements; scope; SIPOC; process mapping levels and symbols [slides 42–105]
14:15–17:00	165 min	Hands-on (core): Activity 2: Lean, Six Sigma, Waste, Voice of Customer, and Value; Activity 3: SIPOC, Process Mapping, Handoffs, and SME Support; Activity 5: DMAIC Overview, Problem Statement, Scope, and Stakeholders. Elective (time permitting): Activity 4: Elective — PDCA Small Improvement Project Charter; Activity 11: Elective — Affinity Diagram and Kano Analysis
17:00–17:30	30 min	Day 1 recap and Q&A

Total instructional time: 450 minutes (7.5 hours) — excludes the 30-minute lunch break.

Day 2 — Measure, Analyse, Improve and Control

Time	Duration	Topic / Activity
9:30–9:50	20 min	Day 1 recap and Day 2 overview
9:50–11:50	120 min	DMAIC · MEASURE — the eight wastes (DOWNTIME); value-added analysis; types of data; data collection plans

		and operational definitions; check sheets; sampling; yield, DPU, DPO, DPMO; sigma level; COPQ [slides 106–143]
11:50–12:20	30 min	Lunch break
12:20–14:10	110 min	Hands-on (core): Activity 6: Data Collection, KPIs, Check Sheets, and Basic Metrics. Elective (time permitting): Activity 12: Elective — Value Stream Map and Takt Time
14:10–15:20	70 min	DMAIC · ANALYZE — common vs special cause variation; Pareto principle and charts; run charts and pattern rules; 5 Whys; Fishbone (5M); brainstorming and multi-voting; the interactive problem-solving toolkit [slides 144–187]
15:20–16:30	70 min	Hands-on (core): Activity 7: Pareto, Run Charts, Variation, Yield, DPU, and DPMO; Activity 8: Root Cause Analysis with 5 Whys, Fishbone, and Evidence using the collaborative Pareto, 5 Whys, Fishbone and NovaSPC tools
16:30–16:50	20 min	DMAIC · IMPROVE — solution generation and selection; 5S; poka-yoke; standard work; Kaizen; piloting [slides 188–221]. Hands-on (core): Activity 9: Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen. Elective: Activity 13: Elective — Solution Selection Matrix, Benchmarking, and FMEA
16:50–17:10	20 min	DMAIC · CONTROL — control plans; process control; visual management; SOPs; team huddles; Gemba; A3 and handover [slides 222–253]. Hands-on (core): Activity 10: Control Plan, A3 Summary, Handover, and Certification Readiness. Elective: Activity 14: Elective — Descriptive Statistics and Implementation Planning
17:10–17:30	20 min	Course recap, key takeaways and next steps [slides 254–261]

Total instructional time: 450 minutes (7.5 hours) — excludes the 30-minute lunch break.

Activity Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Activities
FOUNDATIONS: Six Sigma Foundations	15%	Activity 1
DEFINE: Define — Scope the Problem	25%	Activity 2, Activity 3, Activity 4 (elective), Activity 5, Activity 11 (elective)
MEASURE: Measure — Quantify Performance	25%	Activity 6, Activity 12 (elective)
ANALYZE: Analyze — Find the Root Cause	20%	Activity 7, Activity 8
IMPROVE: Improve — Fix the Cause	10%	Activity 9, Activity 13 (elective)
CONTROL: Control — Hold the Gain	5%	Activity 10, Activity 14 (elective)